

Zac Tawfick

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EDUCATION

Millersville University of Pennsylvania

Millersville, PA

B.S. Computer Science, Minor in Mathematics

December 2026

- **GPA:** 3.99
- **Relevant Coursework:** Compiler Construction, Parallel Programming, Operating Systems, Computer Architecture, Computer Graphics, Software Engineering, Data Structures & Algorithms, Database Systems
- **Competitive Programming:** 2nd place at 2024 CCSC Eastern, 3rd place at 2023 ICPC Division 2

EXPERIENCE

Software Engineer Intern

May 2026 – August 2026

Netflix

Los Gatos, CA

- Accelerated JVM-based ML inference services by embedding Metaflow functions through GraalPy, eliminating an external CPython service, improving p99 latency by 33%, and reducing engineering overhead.
- Contributed POSIX system call support and bug fixes to the open source GraalPy Community Edition, improving CPython compatibility and enabling shared-memory IPC for Python applications.
- Optimized GraalPy JIT performance by 18% by identifying runtime bottlenecks using PyPerformance benchmarking, Linux perf, FlameGraphs, Async Profiler, and eBPF tracing.

Linux Kernel Researcher

January 2026 – Present

Millersville University of Pennsylvania

Millersville, PA

- Implemented the EEVDF scheduling policy in the Linux kernel DRM subsystem, improving fairness while maintaining low scheduling latency for GPU workload dispatch.
- Simplified the DRM scheduler by replacing multiple priority-specific runqueues with a unified scheduling structure and adopting the kernel kref API for safe object lifetime management.
- Validated scheduler behavior using automated kernel builds and custom Vulkan workloads, analyzing fairness, latency, and GPU execution patterns with AMD GPU Top.

Software Developer Intern

May 2025 – August 2025

Gateway Ticketing Systems

Gilbertsville, PA

- Built a C#/.NET desktop application for Windows Service orchestration and configuration, providing operators with a centralized interface for managing service deployment and runtime behavior.
- Implemented performant backend components and service-management workflows, emphasizing reliability, maintainability, and efficient interaction with Windows system services.

Full Stack Developer Intern

December 2024 – January 2025

Nature Atlas Herbarium

Millersville, PA

- Engineered a multi-part upload pipeline for high-volume observation ingestion, improving request handling and data throughput.
- Built backend services for structured geospatial data storage and retrieval using MySQL and Google Maps API.

PROJECTS

Fluid Simulator | C++, OpenMP, CUDA, Linux perf

- Developed a real-time fluid simulation engine in C++ based on the Navier–Stokes equations for physically accurate fluid dynamics.
- Identified hot functions through Linux perf profiling to guide optimization and parallelization of compute-intensive simulation kernels.
- Accelerated simulation performance by over 300% through multithreaded OpenMP parallelization and CUDA GPU offloading of compute-heavy grid updates.

Memory Allocator | C, POSIX system calls, heap checking, gprof

- Implemented a high-performance dynamic memory allocator in C featuring an explicit free list, block splitting, coalescing, and 16-byte alignment, optimizing fragmentation and throughput under trace-driven benchmarking.

TECHNICAL SKILLS

Languages: C/C++, Java, Rust, Python, x86 Assembly, C#, PostgreSQL

Tools & Frameworks: CUDA, Pthreads, MPI, OpenMP, Linux/Unix, Git, eBPF, FlameGraphs, GDB